



METHODIST UNIVERSITY GHANA

FACULTY OF SCIENCE
DEPARTMENT OF NURSING & MIDWIFERY

END OF SECOND SEMESTER EXAMINATIONS, 2023/2024 ACADEMIC YEAR
BSc GENERAL NURSING, LEVEL 200

MNUR 204: MEDICAL NURSING I
(Examiner: Mad. Nadia Abbey)

04.30 Friday, May 10, 2024

TIME ALLOWED: 2 HOURS

INSTRUCTIONS: ANSWER ALL QUESTIONS. ANSWER SECTION A ON THE QUESTION PAPER AND SECTION B IN THE ANSWER BOOKLET PROVIDED.

SECTION A [120 Marks]

Read the questions carefully and select the best choice for each question. Circle the letter corresponding to your choice

1. Which of the following is NOT a symptom of meningitis?
 - a. Photophobia
 - b. Tinnitus
 - c. Headache
 - d. Neck stiffness
2. Which diagnostic test is most commonly used to confirm the presence of bacterial meningitis?
 - a. Blood culture
 - b. Lumbar puncture
 - c. CT scan
 - d. MRI
3. The nurse is assessing the client diagnosed with bacterial meningitis. Which clinical manifestations would support the diagnosis of bacterial meningitis?
 - a. Positive Kernig's sign and nuchal rigidity.
 - b. Negative Trousseau's sign and nystagmus.
 - c. Positive Babinski's sign and peripheral paraesthesia.
 - d. Negative Chvostek's sign and facial tingling.
4. What is the classic triad of symptoms seen in bacterial meningitis?
 - a. Fever, headache, and confusion
 - b. Neck stiffness, photophobia, and altered mental status
 - c. Vomiting, irritability, and lethargy
 - d. Seizures, rash, and diarrhea
5. Which of the following is a potential complication of untreated bacterial meningitis?
 - a. Nephritis
 - b. Liver failure
 - c. Osteoporosis

- d. Hearing loss
- 6. A 32-year-old female is about to deliver a baby. The patient is HIV-positive and has been taking antiretroviral therapy during the pregnancy. What steps can be taken to help prevent transmitting the virus to the baby after birth?
 - a. Substitute formula for breastfeeding.
 - b. Administer antiretroviral treatment to the newborn for 2 weeks after birth.
 - c. Avoid kissing and hugging the newborn.
 - d. Stop taking antiretroviral therapy for 2 months postpartum
- 7. Madam Esthera patient with HIV is prescribed to start antiretroviral therapy. The nurse is providing education about these medications. Which statement below by the patient indicates they need re-education on these medications.
 - a. Drug resistance is likely to develop if I'm non-compliant with my medications.
 - b. If I take these medications as prescribed my viral load will become undetectable, and I have a low risk of transmitting the virus to others.
 - c. I currently take a medication called St. John's Wort to treat depression.
 - d. This therapy does not cure me from HIV but helps me live a healthier and longer life.
- 8. Mr. Appiah, who is diagnosed with bipolar disorder has been drinking copious amounts of water and voiding frequently. The patient is experiencing muscle cramps, twitching, and is reporting dizziness. The nurse checks lab work for:
 - a. EEG results
 - b. Complete blood count results, particularly the platelets
 - c. Electrolytes, particularly the serum sodium
 - d. Urine analysis, particularly for the presence of white blood cells
- 9. The balance of anions and cations as it occurs across cell membranes is known as:
 - a. Osmotic activity
 - b. Electrical neutrality
 - c. Electrical stability
 - d. Sodium-potassium pump
- 10. Opoku, an 85-year-old patient with a feeding tube, has been experiencing severe watery stool. The patient is lethargic and has poor skin turgor, a pulse of 120, and hyperactive reflexes. Nursing interventions would include:
 - a. Administering sedatives.
 - b. Applying wrist restraints to avoid displacement of the feeding tube.
 - c. Measuring and recording intake and output and daily weights
 - d. Administering salt tablets and monitoring hypertonic parenteral solutions.
- 11. Bicarbonate is lost during which of the following clinical conditions?
 - a. Diarrhea
 - b. Diuresis
 - c. Diaphoresis
 - d. Vomiting
- 12. Which of the following clinical conditions exacerbates electrolyte excretion?
 - a. Nasogastric feedings
 - b. Immobility from fractures
 - c. Use of surgical drains
 - d. Chronic water drinking
- 13. Which of the following electrolytes are lost as a result of vomiting?
 - a. Bicarbonate and calcium
 - b. Sodium and hydrogen

- c. Sodium and potassium
- d. Hydrogen and potassium
- 14. Which of the following electrolyte imbalances can be life-threatening due to its potential to cause cardiac arrhythmias?
 - a. Hyperkalemia
 - b. Hyponatremia
 - c. Hypomagnesemia
 - d. Hypokalemia
- 15. Which of the following is not an appropriate nursing intervention for a patient with hypercalcemia?
 - a. Administering calcitonin
 - b. Administering loop diuretics
 - c. Encouraging ambulation
 - d. Administering calcium gluconate
- 16. Which electrolyte imbalance is characterized by muscle twitching, tingling sensations, and cardiac arrhythmias?
 - a. Hyponatremia
 - b. Hypercalcemia
 - c. Hypocalcemia
 - d. Hyponatremia
- 17. Which of the following intravenous solutions would be appropriate for a patient with severe hyponatremia secondary to syndrome of inappropriate antidiuretic hormone (SIADH)?
 - a. Hypotonic solution
 - b. Hypertonic solution
 - c. Isotonic solution
 - d. Normotonic solution
- 18. To determine if a patient's respiratory system is functioning, the nurse would assess which of the following parameters:
 - a. Respiratory rate
 - b. Pulse
 - c. Arterial blood gas
 - d. Pulse oximetry
- 19. Match the acid-base status of the following blood samples to the disorders in the given choices. (PaCO₂ values are in mm Hg and bicarbonate values in mmol/l). PH 7.57, PaCO₂ 22, HCO₃⁻ 17
 - a. Respiratory Acidosis, Partially Compensated
 - b. Respiratory Alkalosis, Uncompensated
 - c. Metabolic Acidosis, Partially Compensated
 - d. Respiratory Alkalosis, Partially Compensated
- 20. Match the acid-base status of the following blood samples to the disorders in the given choices. (PaCO₂ values are in mm Hg and bicarbonate values in mmol/l)
 - a. Respiratory Acidosis
 - b. Metabolic Acidosis
 - c. Respiratory Alkalosis
 - d. Normal
- 21. A mother is admitted to the emergency department following complaints of fever and chills. The nurse on duty took her vital signs and noted the following: Temp = 37.8 °C; apical pulse = 95; respiration = 20 and deep. Measurement of arterial

blood gas shows pH 7.37, PaO₂ 90 mm Hg, PaCO₂ 40 mm Hg, and HCO₃ 24 mmol/L. What is your assessment?

- a. Hyperthermia
- b. Hyperthermia and Respiratory Alkalosis
- c. Hypothermia
- d. Hypothermia and Respiratory Alkalosis

22. Mr William is a 64-year-old widower with a history of chronic obstructive pulmonary disease and was rushed to the emergency department with increasing shortness of breath, pyrexia, and a productive cough with yellow-green sputum. He has difficulty communicating because of his inability to complete a sentence. One of his sons, Jacob, says he has been unwell for three days. Upon examination, crackles and wheezes can be heard in the lower lobes; he has tachycardia and a bounding pulse. Measurement of arterial blood gas shows pH 7.3, PaCO₂ 68 mm Hg, HCO₃ 28 mmol/L, and PaO₂ 60 mm Hg. How would you interpret this?

- a. Respiratory Acidosis, Uncompensated
- b. Metabolic Alkalosis, Uncompensated
- c. Respiratory acidosis, partially compensated
- d. Metabolic Acidosis, Partially Compensated

23. Abena, an elementary student, was rushed to the hospital due to vomiting and a decreased level of consciousness. The patient displays slow and deep (Kussmaul breathing), and he is lethargic and irritable in response to stimulation. He appears to be dehydrated—his eyes are sunken and mucous membranes are dry—and he has a two-week history of polydipsia, polyuria, and weight loss. Measurement of arterial blood gas shows pH 7.0, PaO₂ 90 mm Hg, PaCO₂ 23 mm Hg, and HCO₃ 12 mmol/L; other results are Na⁺ 126 mmol/L, K⁺ 5 mmol/L, and Cl⁻ 95 mmol/L. What is your assessment?

- a. Respiratory Acidosis, Uncompensated
- b. Respiratory Acidosis, Partially Compensated
- c. Metabolic Alkalosis, Uncompensated
- d. Metabolic Acidosis, Partially, Compensated

24. Mrs Mensah a cigarette vendor was brought to the emergency department of a hospital after she fell into the ground and hurt her left leg. She is noted to be tachycardic and tachypnoeic. Painkillers were carried out to lessen her pain. Suddenly, she started complaining that she is still in pain and now experiencing muscle cramps, tingling, and paraesthesia. Measurement of arterial blood gas reveals pH 7.6, PaO₂ 120 mm Hg, PaCO₂ 31 mm Hg, and HCO₃ 25 mmol/L. What does this mean?

- a. Respiratory Acidosis, Partially Compensated
- b. Metabolic Alkalosis, Uncompensated
- c. Metabolic Alkalosis, Partially Compensated
- d. Respiratory Alkalosis, Uncompensated

25. Mr. Mohammed a 49-year-old company driver is found at the scene of an automobile accident in a state of emotional distress. He tells the paramedics that he feels dizzy, tingling in his fingertips, and does not remember what happened to his car. Respiratory rate is rapid at 34/minute. Which primary acid-base disturbance is the young man at risk for if medical attention is not provided?

- a. Respiratory Alkalosis
- b. Respiratory Acidosis
- c. Metabolic Acidosis
- d. Metabolic Alkalosis

26. In a patient undergoing surgery, it was vital to aspirate the contents of the upper gastrointestinal tract. After the operation, the following values were acquired from an arterial blood sample: pH 7.55, PCO₂ 52 mm Hg and HCO₃⁻ 40 mmol/l. What is the underlying disorder?
 - a. Respiratory Acidosis
 - b. Respiratory Alkalosis
 - c. Metabolic Alkalosis
 - d. Metabolic Acidosis
27. Marry Ann, who is drinking beer at a party, falls and hits her head on the ground. Her friend Bertina dials "199" because Maryann is unconscious, depressed ventilation (shallow and slow respirations), rapid heart rate, and is profusely bleeding from both ears. Which primary acid-base imbalance is Maryann at risk for if medical attention is not provided?
 - a. Respiratory alkalosis
 - b. Respiratory acidosis
 - c. Metabolic alkalosis
 - d. Metabolic acidosis
28. Hannah a patient who is hospitalized due to vomiting and a decreased level of consciousness displays slow and deep (Kussmaul breathing), and he is lethargic and irritable in response to stimulation. The doctor diagnosed him of having dehydration. Measurement of arterial blood gas shows pH 7.0, PaO₂ 90 mm Hg, PaCO₂ 22 mm Hg, and HCO₃⁻ 14 mmol/L; other results are Na⁺ 120 mmol/L, K⁺ 2.5 mmol/L, and Cl⁻ 95 mmol/L. As a knowledgeable nurse, you know that the normal value for PaCO₂ is:
 - a. 22 mm Hg
 - b. 48 mm Hg
 - c. 38 mm Hg
 - d. 50 mm hg
29. Isaac, a 8-year-old boy, was rushed to the hospital following her mother's complaint that her son has been vomiting, nauseated and has overall weakness. After a series of tests, the nurse notes the laboratory results: potassium: 2.9 mEq. Which primary acid-base imbalance is this boy at risk for if medical intervention is not carried out?
 - a. Respiratory Acidosis
 - b. Respiratory Alkalosis
 - c. Metabolic Acidosis
 - d. Metabolic Alkalosis
30. The nurse would anticipate which of the following ABG results in a client experiencing a prolonged, severe asthma attack?
 - a. Decreased PaCO₂, increased PaO₂, and decreased pH.
 - b. Increased PaCO₂, decreased PaO₂, and decreased pH.
 - c. Increased PaCO₂, increased PaO₂, and increased pH.
 - d. Decreased PaCO₂, decreased PaO₂, and increased pH.
31. Which electrolyte imbalance can lead to cardiac dysrhythmias and ECG changes such as widened QRS complexes and peaked T waves?
 - a. Hyperkalemia
 - b. Hypokalemia
 - c. Hypernatremia
 - d. Hyponatremia

32. Which of the following statements provides the rationale for using a hypotonic solution for a patient with fluid volume deficit?
- a hypotonic solution provides free water to help the kidneys eliminate the solute.
 - A hypotonic solution supplies an excess of sodium and chloride ions.
 - Excessive volumes are recommended in the early postoperative period.
 - A hypotonic solution is used to treat hyponatremia.
33. Oware is admitted with 1,000 ml of diarrhea per day for the last 3 days. An IV of 0.45% NaCl mixed with 5% dextrose is infusing. Which of the following nursing interventions is the most appropriate?
- Get an infusion controller from the central supply.
 - Mix all antibiotics in 0.45% NaCl with 5% dextrose.
 - Check the patient's potassium level and contact the doctor for IV additive orders.
 - Assess the patient for signs of hyperkalemia.
34. Diabetic ketoacidosis, aspirin toxicity, and renal failure are examples of the causes of ____.
- High anion gap metabolic acidosis
 - Normal anion gap metabolic acidosis
 - Low anion gap metabolic acidosis
 - Normal anion gap respiratory acidosis
35. A patient has the following arterial blood gases: HCO₃ 38, pH 7.50, PaCO₂ 50. Which of the following signs may this patient exhibit as a compensatory mechanism?
- Hyperventilation (tachypnea)
 - Hypoventilation (bradypnea)
 - Increased potassium level (hyperkalemia)
 - Constipation
36. A patient states they have been vomiting for the last 4 days. The patient is irritable, weak, and reporting muscle cramping and weakness. On assessment, the patient is experiencing bradypnea with a respiratory rate of 10. The patient has the following ABGs result: HCO₃ 36, pH 7.52, PaCO₂ 48. Which condition below is presenting?
- Metabolic alkalosis partially compensate
 - Metabolic alkalosis fully compensated.
 - Metabolic acidosis partially compensated.
 - Metabolic acidosis not compensated
37. A patient is in high anion gap metabolic acidosis due to diabetic ketoacidosis. Which of the following signs and symptoms would you expect to see in this patient?
- Kussmaul's respirations
 - Glucose 110
 - Hypoventilation
 - Neuro-excitability
38. A patient has the following blood gases: PaCO₂ 25, pH 7.50, HCO₃ 19. Which of the following could NOT be the cause of this condition?
- Anxiety attack.
 - Chronic obstructive pulmonary disease (COPD)
 - Fever
 - Aspirin toxicity

39. Which of the following is not a cause of respiratory acidosis?
- Pulmonary edema
 - Asthma
 - Chronic obstructive pulmonary disease (COPD)
 - Hyperventilation
40. Kwaku is post-operative patient from knee surgery. He has been receiving Morphine 4 mg IV every 2 hours. You notice the patient is exhibiting a respiratory rate of 8 and is extremely drowsy. Which of the following conditions is the patient at risk for?
- Respiratory acidosis
 - Respiratory alkalosis
 - Hypokalemia
 - Metabolic acidosis
41. All are true for renal handling of acids in metabolic acidosis except:
- Hydrogen ion secretion is increase
 - Bicarbonate reabsorption is decreased
 - Urinary acidity is increase
 - Urinary ammonia is increased.
42. Causes of lactic acidosis include all except
- Acute Myocardial infarction
 - Hypoxia
 - Circulatory failure.
 - Infections
43. The healthcare provider is assessing the skin of a patient who is at risk for becoming infected with the human immunodeficiency virus (HIV). Which of the following findings requires immediate follow-up by the healthcare provider?
- Purplish-red raised lesions
 - Ecchymosis on the legs
 - Patches of dry, flaky skin
 - Numerous moles on the chest and back
44. A patient who has been diagnosed with acquired immunodeficiency syndrome (AIDS) develops an oral *Candida* infection. When teaching the patient, the healthcare provider will include which of the following instructions?
- Include plenty of citrus juices in your diet
 - Select foods that are soft or pureed
 - Include hot soups and beverages with each meal
 - Rinse your mouth often with a commercial mouthwash
45. A patient who has been receiving antiretroviral therapy (ART) to manage infection with human immunodeficiency virus (HIV) has an undetectable viral load. How would the healthcare provider interpret this information?
- HIV has been eliminated from the patient's blood
 - ART has been effective in decreasing viral load
 - More tests are needed to determine the effectiveness of ART
 - ART can be discontinued for three months
46. A male patient arrives at the polyclinic clinic reporting the following: painful and swollen testicles, burning with urination, and a yellow-green discharge from the penis. The healthcare provider understands that these symptoms are likely due to which of these problems?
- Stage 2 syphilis
 - Epididymitis

- c. Gonorrhea
 - d. Trichomoniasis
47. A patient diagnosed with trichomoniasis is being treated with metronidazole (Flagyl). When teaching the patient about this medication, which of the following will the healthcare provider include?
- a. Call our office if you experience any tendon pain or tenderness
 - b. Report the occurrence of pain in your upper abdomen immediately
 - c. You should avoid milk or dairy products during therapy
 - d. Do not drink alcohol while you are taking this medication
48. What is pitting edema?
- a. Edema that occurs only in the extremities
 - b. Edema characterized by the retention of fluid causing an indentation when pressure is applied
 - c. Edema that occurs due to allergic reactions
 - d. Edema that resolves on its own without treatment
49. Which lifestyle change can help reduce edema?
- a. Decreasing sodium intake
 - b. Increasing caffeine consumption
 - c. Smoking more cigarettes
 - d. Avoiding exercise
50. A client has an order to have radial ABG drawn. Before drawing the sample, a nurse occludes the:
- a. Brachial and radial arteries, and then releases them and observes the circulation of the hand.
 - b. Radial and ulnar arteries, releases one, evaluates the color of the hand, and repeats the process with the other artery.
 - c. Radial artery and observes for color changes in the affected hand.
 - d. Ulnar artery and observes for color changes in the affected hand
51. Which of the following ABG abnormalities should the nurse anticipate in a client with advanced COPD?
- a. Increased PaCO₂
 - b. Increased PaO₂
 - c. Increased pH
 - d. Increased oxygen saturation
52. What signs and symptoms of fluid volume deficit are not likely to present in older adults (geriatric patient populations)?
- a. Sluggish capillary refill and confusion
 - b. Confusion and weight loss
 - c. Thirst and sluggish capillary refill
 - d. Decreased skin turgor and thirst
53. A patient is receiving treatment for fluid volume deficit (hypertonic dehydration). Which of this will the nurse not implement in the patient's plan of care?
- a. Daily weights
 - b. Administer hypotonic IV solutions per physician's order
 - c. Increase sodium intake
 - d. Strict intake and output measurements
54. What signs and symptoms could not be present with fluid volume deficit in an adult patient?
- a. Weight loss and confusion
 - b. Pitting edema and increased skin turgor